Lesson 8: Developing a Complete Student Management System

Focus: Applying classes and objects to solve a practical problem.

Combine all the functionality developed in the previous lessons into one working application.

Exercise 8 – Final Practical Project

Develop a Student Management System with the following menu:

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       STUDENT MANAGEMENT SYSTEM
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1. Register a new student
2. Display all students
3. Search for a student
4. Update student information
5. Enter or update examination marks
6. Display student results
7. Display class performance report
8. Remove a student
9. Exit

Enter your choice:
```

System requirements

Your application must:

1. Use a `Student` class to represent individual students.
2. Use a constructor to initialise student information.
3. Use appropriate attributes and methods.
4. Apply encapsulation to protect important student data.
5. Validate all marks entered by users.
6. Store multiple Student objects in a list or array.
7. Allow users to register new students.
8. Allow users to search for students using admission numbers.
9. Allow users to update student information and examination marks.
10. Calculate totals, averages and grades.
11. Generate a class performance report.
12. Allow users to delete student records.
13. Continue displaying the menu until the user chooses to exit.

Additional challenge

Enhance the application by introducing a second class called `Subject`.

The `Subject` class should store:

- Subject code.
- Subject name.
- Subject teacher.

Modify your system so that students can be associated with the subjects they study.